

Mathang Peddi

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EXPERIENCE

iTradeNetwork | *Software Engineer* | Pleasanton, California

Jan 2026 - Present

- Developed an **AI-driven file ingestion agent** to parse documents and generate purchase orders in **OMS**, reducing manual effort and improving processing time by 45%
- Designed **Spring Boot** microservices with **REST APIs** and **Hibernate ORM**, reducing API latency by **40%** using optimized indexing and connection pooling

The Home Depot | *Software Engineer Intern* | Atlanta, Georgia

Aug 2025 - Dec 2025

- Built an LLM evaluation engine using **LangSmith**, **OpenRAG-Eval**, and Azure OpenAI, processing 1,500+ queries per run and detecting grounding, retrieval, and hallucination issues with **92%** accuracy
- Integrated evaluation metrics like **UMBRELA**, **AutoNugget**, Citation Support, and Precision/Recall to assess retrieval relevance, **context alignment**, and citation accuracy
- Utilized GCP's **BigQuery** to streamline and optimize large-scale data workflows, employing **Vertex AI**, cutting data-to-insight time from 5 hours to under 30 mins
- Leveraged AI coding tools like **Cursor**, **Copilot** to accelerate backend development via code generation and debugging, reducing development time by **60%**

Juniper Networks | *Cloud Engineer Intern* | Sunnyvale, California

May 2025 - Aug 2025

- Architected **Golang** backend APIs powering 20+ application and security dashboard widgets, using custom **Elasticsearch** queries to process 300GB+ telemetry data in under 2s
- Refactored **40+ Go microservices**, rebuilding APIs with **gRPC** to maintain FIPS 140 security standards, fix vulnerabilities, and deliver reproducible **Bazel-built artifacts**
- Implemented **data partitioning and sharding** strategies to scale distributed services, enabling horizontal scaling across multiple nodes and supporting a **3x increase in data volume** without performance degradation
- Streamlined audit logging using **AWS CloudTrail**, **CloudWatch**, using KMS, and ensuring SOC2/GDPR compliance and traceability of user and admin actions

Evidence Based Cybersecurity Lab | *Research Analyst* | Atlanta, Georgia

Jan 2024 - May 2025

- Developed a Natural Language to SQL tool using **Google Generative AI** embeddings and **Chroma** vector store, enabling non-technical users to query databases in plain English with dynamic table selection and contextual memory.
- Incorporated **OpenAI API** to create a tool that **scans research papers**, identifies methodological flaws, raising peer-review acceptance rates by 15% across conference submissions
- Integrated **GitHub Actions** and a real-time dashboard to automate PR analysis, visualize flagged issues, and create a production-ready system for seamless developer adoption and team-wide collaboration
- Pioneered a **Named Entity Recognition (NER)** model to identify key entities such as contract names, dates, and organizations within user queries, significantly enhancing the precision of search results

Citrix Systems | *Software Engineer* | Bangalore, India

Jan 2022 - Jan 2024

- Achieved **\$750K+** in revenue growth by engineering the Activity Manager module - a **centralized session monitoring system**, incorporated into **Citrix Workspace (WSUI)**, now adopted by **3000+** global customers
- Implemented a WSUI scheduling feature for automated desktop lifecycle management, **reducing cloud VM costs** and winning **1st place** at a company-wide hackathon
- Created production-grade WSUI modules with **Next.js**, **Vue.js**, **Tailwind CSS**, **Node.js**, **MongoDB**, **MySQL**, enabling continuous updates, and event-driven architecture for seamless backend integration
- Optimized WSUI performance with **Redux Toolkit**, **React Query**, memoization, lazy loading, debouncing, service worker, reducing component render time by **50%** and improving perceived page load speed to **0.5s (80% faster)**

PROJECTS

Agentic AI for Dependency Migration

(Neo4j, LLMs, LangGraph, Docker, CI/CD)

- Developed an agentic AI platform that autonomously audits and upgrades module dependencies, utilizing **graph-based** program analysis and refactoring to reduce vulnerabilities by **40%**
- Implemented a **multi-agent pipeline** for dependency scanning, automated fixes, unit testing, and pull request generation, scaling across **200+ repositories** and achieving a 92% automated merge success rate

Real-Time Utility Consumption Platform

(Next.js, Node.js, Kafka, Docker, RabbitMQ, AWS)

- Developed an analytics dashboard using **Kafka event streams** to track electricity, water, and gas usage, providing **consumption insights and predictive billing** through dynamic trend visualizations
- Deployed containerized microservices on Kubernetes with MongoDB persistence, **Redis caching** and asynchronous message queues, processing **100k+ meter events/min** with fault-tolerant rollouts via CI/CD pipelines on **AWS**

TECHNICAL SKILLS

Programming Languages: JavaScript/TypeScript, C++, Python, Java, SQL/NoSQL

Web Dev: HTML, CSS, React, Next JS, Node, MongoDB, FastAPI, gRPC, AWS, Spring Boot, RTL, SDL C

ML & AI: Gen AI, Langchain, Numpy, Pandas, Sci-kit learn, NLTK, Keras, Tensorflow

Additional: Agile, CI/CD, Jenkins, Git, Docker, Kubernetes, OOPS, Shell Scripting(Linux/Unix)

Publications: Co-authored a chapter in the book '**Intelligent Systems and Machine Learning for Industry**', detailing an end-to-end speech recognition model for recognizing different native accents of English

EDUCATION

Georgia State University | *Master of Science in Computer Science*

Jan 2024 – Dec 2025

Courses: Data Structures, Distributed Systems, Text Processing and LLMs, Database Systems